Simulation task

1. Simulation of MIMO channel capacity under antenna number, SNR, bandwidth, fading model, and multiplexing order (BL4)

Code

```
clc;

clear;

close all;


%% Parameters

SNR_dB = 0:5:30;        % SNR range (dB)

Nt_list = [2 4 8];      % Number of transmit antennas

Nr = 4;                 % Number of receive antennas

Bandwidth = [5 10 20 40]; % MHz

K = 5;                  % Rician K-factor


%% ---------------------------------------------------

%% 1. Capacity vs SNR for different antenna numbers

%% ---------------------------------------------------


figure;


for a = 1:length(Nt_list)


   Nt = Nt_list(a);

   Capacity = zeros(size(SNR_dB));


   for i = 1:length(SNR_dB)
```

```
        snr = 10^(SNR_dB(i)/10);

        % Rayleigh channel
        H = (randn(Nr,Nt)+1j*randn(Nr,Nt))/sqrt(2);

        Capacity(i) = real(log2(det(eye(Nr) + (snr/Nt)*(H*H'))));

    end

    plot(SNR_dB,Capacity,'LineWidth',2);
    hold on;

end

grid on
xlabel('SNR (dB)')
ylabel('Capacity (bits/s/Hz)')
title('MIMO Capacity vs SNR')
legend('2x4','4x4','8x4','Location','northwest')

%% ---------------------------------------------------
%% 2. Capacity vs Bandwidth
%% ---------------------------------------------------

CapacityBW = zeros(size(Bandwidth));

Nt = 4;
```

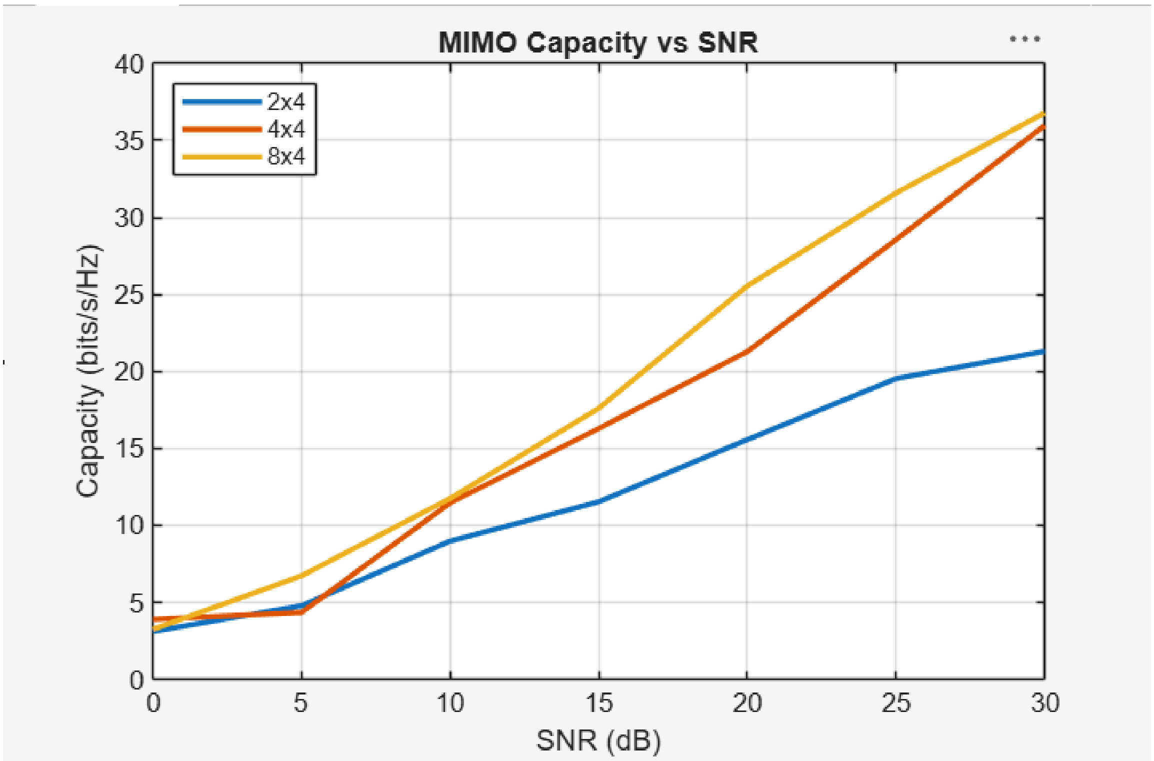
```
snr = 10^(20/10);

for i = 1:length(Bandwidth)

    H = (randn(Nr,Nt)+1j*randn(Nr,Nt))/sqrt(2);

    C = real(log2(det(eye(Nr)+(snr/Nt)*(H*H'))));

    CapacityBW(i)=Bandwidth(i)*1e6*C/1e6;

end

figure
plot(Bandwidth,CapacityBW,'o-','LineWidth',2)
grid on
xlabel('Bandwidth (MHz)')
ylabel('Capacity (Mbits/s)')
title('Effect of Bandwidth')

%% ---------------------------------------------------
%% 3. Rayleigh vs Rician Fading
%% ---------------------------------------------------

CapacityRay = zeros(size(SNR_dB));
CapacityRic = zeros(size(SNR_dB));

Nt = 4;
```

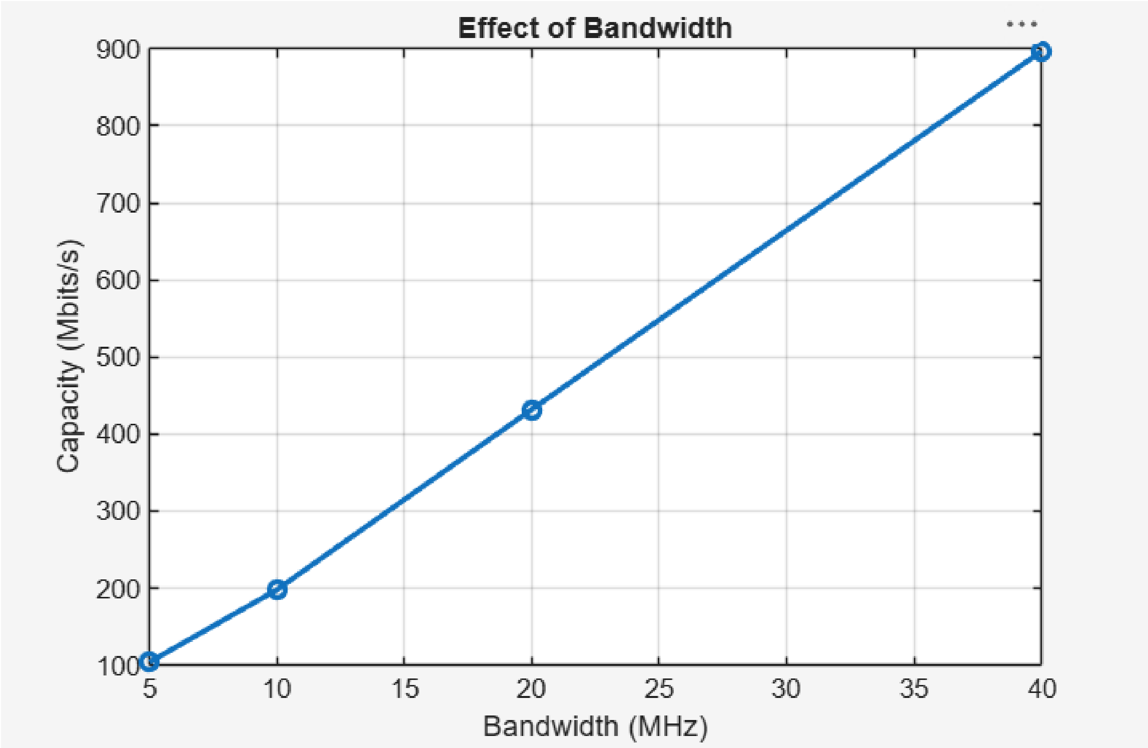
```
for i = 1:length(SNR_dB)

    snr = 10^(SNR_dB(i)/10);

    %% Rayleigh Channel
    Hr = (randn(Nr,Nt)+1j*randn(Nr,Nt))/sqrt(2);

    %% Rician Channel
    Hric = sqrt(K/(K+1))*ones(Nr,Nt) + ...
           sqrt(1/(K+1))*((randn(Nr,Nt)+1j*randn(Nr,Nt))/sqrt(2));

    CapacityRay(i)=real(log2(det(eye(Nr)+(snr/Nt)*(Hr*Hr'))));

    CapacityRic(i)=real(log2(det(eye(Nr)+(snr/Nt)*(Hric*Hric'))));

end

figure
plot(SNR_dB,CapacityRay,'LineWidth',2)
hold on
plot(SNR_dB,CapacityRic,'--','LineWidth',2)
grid on
xlabel('SNR (dB)')
ylabel('Capacity (bits/s/Hz)')
title('Rayleigh vs Rician Fading')
legend('Rayleigh','Rician')

%% ---------------------------------------------------
```

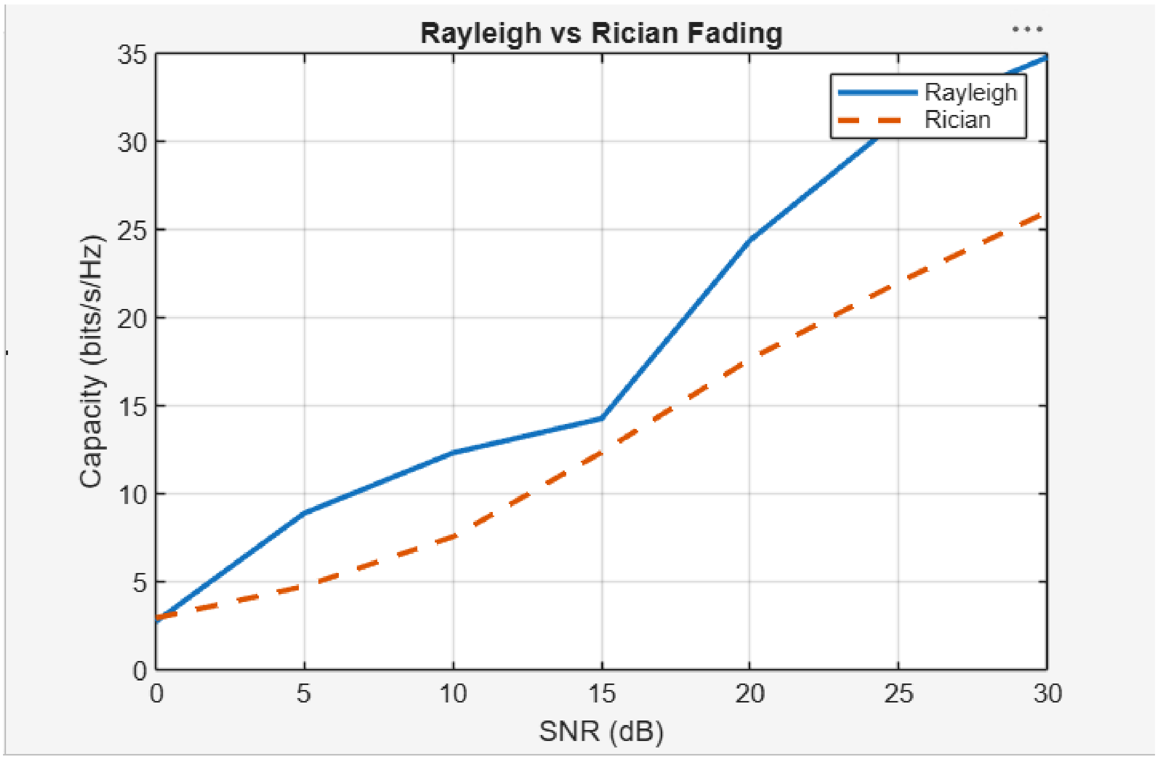
```
%% 4. Capacity vs Multiplexing Order
%% ----------------------------------------------------

Streams = 1:4;

CapacityStream = zeros(size(Streams));

snr = 10^(20/10);

for i = 1:length(Streams)

    Ns = Streams(i);

    H = (randn(Nr,Ns)+1j*randn(Nr,Ns))/sqrt(2);

    CapacityStream(i)=real(log2(det(eye(Nr)+(snr/Ns)*(H*H'))));

end

figure
bar(Streams,CapacityStream)
grid on
xlabel('Spatial Multiplexing Order')
ylabel('Capacity (bits/s/Hz)')
title('Effect of Multiplexing Order')

%% ----------------------------------------------------
%% Display Results
```

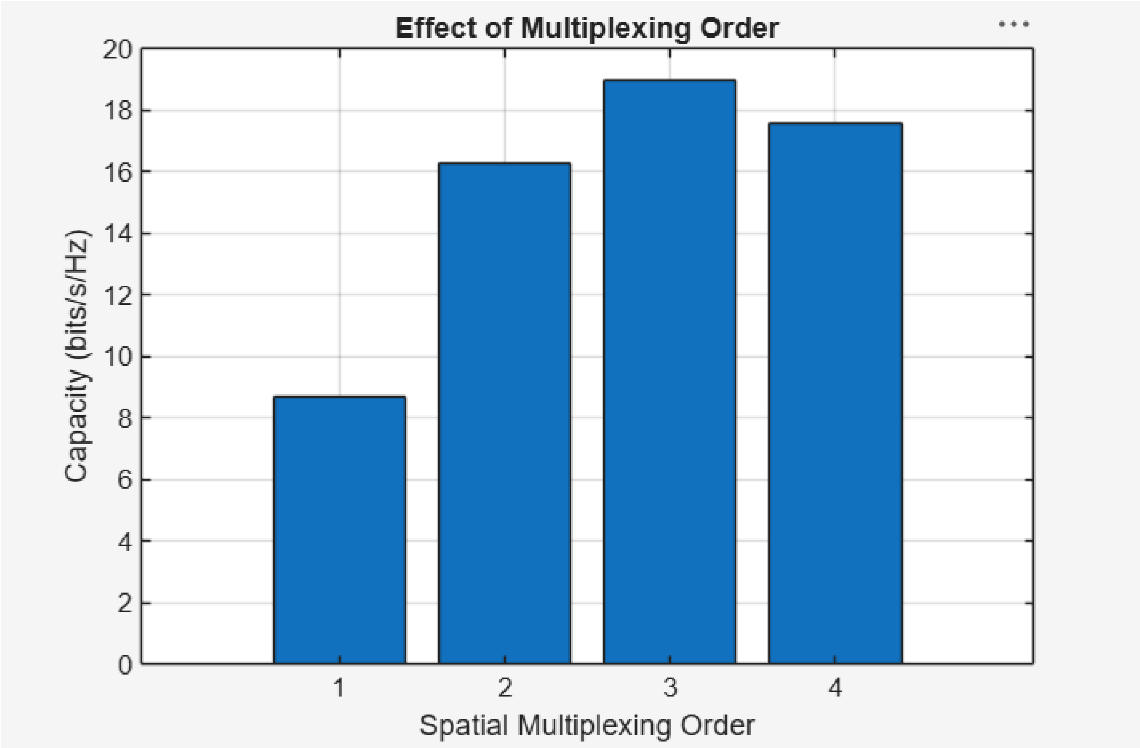
```
%% ----------------------------------------------------

disp('------------------------------------------')
disp('MIMO CHANNEL CAPACITY SIMULATION')
disp('------------------------------------------')

fprintf('Receive Antennas      : %d\n',Nr);
fprintf('Transmit Antennas     : %d, %d, %d\n',Nt_list);
fprintf('SNR Range             : 0 to 30 dB\n');
fprintf('Bandwidth Tested      : 5, 10, 20, 40 MHz\n');
fprintf('Fading Models         : Rayleigh and Rician\n');
fprintf('Multiplexing Streams  : 1 to 4\n');

disp('Simulation Completed Successfully.')
```

output




Effect of Bandwidth
Capacity (Mbits/s)
900
800
700
600
500
400
300
200
100
5
10
15
20
25
30
35
40
Bandwidth (MHz)


Rayleigh vs Rician Fading
Rayleigh
Rician
Capacity (bits/s/Hz)
35
30
25
20
15
10
5
0
0
5
10
15
20
25
30
SNR (dB)


Effect of Multiplexing Order
Capacity (bits/s/Hz)
20
18
16
14
12
10
8
6
4
2
0
1
2
3
4
Spatial Multiplexing Order